

INTRODUCTION

The online gaming industry has experienced exponential growth, with platforms like Steam connecting millions of players worldwide. This project initially began as an exploration of general player behavior but evolved to focus specifically on predicting high-engagement players - a critical business need for game developers. Understanding what drives players to remain engaged has significant implications for game design, monetization strategies, and player retention.

The project's scope evolved from a broad exploration of gaming habits to a focused analysis of engagement prediction, as we recognized that player retention is the cornerstone of sustainable gaming businesses. We also refined our definition of "high engagement" throughout the project, moving from subjective criteria to data-driven metrics based on session frequency, duration, and in-game progression.

Our analysis progressed through three key phases: defining the problem and understanding the data, preprocessing and exploratory data analysis, and finally, model development and evaluation. At each stage, our understanding of player engagement deepened, ultimately allowing us to identify the most significant predictors of high-value players.

PROBLEM DEFINITION AND OBJECTIVES

In the highly competitive gaming market, player retention represents a major challenge. While acquiring new players is important, keeping them engaged over time is more cost-effective and valuable for sustainable growth. Our project addresses this by analyzing player behavior to predict engagement levels.

Specifically, we aim to:

- Identify key factors that drive sustained player engagement: We seek to determine which behavioral, demographic, and game-specific factors most strongly correlate with high engagement. This includes analyzing how playtime patterns, achievement progression, and session frequency contribute to overall retention.
- Characterize high-value players who make in-game purchases: Beyond engagement, we aim to understand the specific attributes and behaviors that distinguish players who financially invest in games. This helps developers optimize monetization strategies without compromising player experience.
- Discover cross-genre patterns affecting game ownership and retention: By analyzing behaviors across different game types, we can uncover universal engagement principles as well as genre-specific considerations that affect player commitment.

These insights enable game developers to implement data-driven strategies for:

- Designing more effective matchmaking systems that pair players with compatible skill levels and play styles
- Refining game mechanics to capitalize on elements that drive engagement
- Creating personalized reward systems that respond to individual player behaviors
- Developing targeted retention strategies for players showing early signs of disengagement
- Balancing monetization approaches that enhance rather than detract from player experience

DATA SOURCES AND DESCRIPTION

Our analysis leverages two complementary datasets:

- "Predict Online Gaming Behavior Dataset": A refined dataset containing player behavior metrics. It captures the essential variables needed for engagement prediction, including demographics, gameplay patterns, and achievement statistics. This dataset serves as our primary analytical foundation.
- "Steam Games Dataset": A comprehensive collection of game attributes from one of the largest digital distribution platforms. It includes rich metadata about game attributes, historical player statistics, rating patterns, and pricing information. We use this to provide contextual information about game characteristics and industry trends.

Features: The combined dataset offers 13 key attributes across three categories

- Player Demographics:
 - o PlayerID: Unique identifier for each player.
 - o Age: Age of the player.
 - o Gender: Gender of the player.
 - o Location: Geographic location of the player.
- Game-Specific Attributes:
 - o GameGenre: Genre of the game the player is engaged in (e.g., RPG, Strategy, Action).
 - o GameDifficulty: Difficulty level of the game, categorized as Easy, Medium, or Hard.
 - o PlayerLevel: Current level of the player in the game.
 - o AchievementsUnlocked: Number of achievements unlocked by the player.
- Engagement Metrics:
 - o PlayTimeHours: Average hours spent playing per session.
 - o InGamePurchases: Indicates whether the player makes in-game purchases (0 = No, 1 = Yes).
 - o SessionsPerWeek: Number of gaming sessions per week.
 - o AvgSessionDurationMinutes: Average duration of each gaming session in minutes.
- Target Variable:
 - o EngagementLevel: Categorizes player engagement levels into three classes: 'High' and 'Low-Medium' reflecting player retention and activity.

Our target variable classification was simplified during the project from an initial three-level categorization (High, Medium, Low) to a binary classification (High, Low-Medium). This modification improved model performance and business relevance by creating a clearer distinction between highly engaged players and those with moderate to low engagement.

EXPLORATORY DATA ANALYSIS AND INSIGHTS

We wanted to check the Class Distribution and Balance of our dataset. Upon our initial exploration revealed an imbalanced distribution of engagement levels, with 74.2% of players classified as "Low-Medium" engagement and only 25.8% showing "High" engagement. This represents a significant class imbalance that influenced our modeling approach.

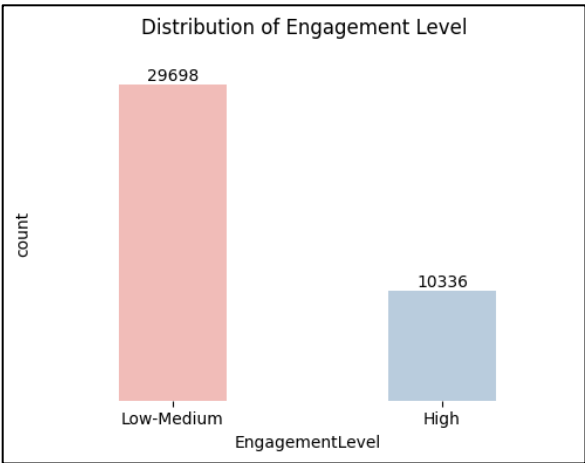


Fig.1

Fig. 1 shows the distribution of the two classes for our target variable. The imbalance reflects real-world gaming ecosystems where highly engaged players form a valuable minority. We addressed this methodologically by using evaluation metrics resistant to class imbalance (like F1-score) and techniques that prevent models from simply predicting the majority class.

We then wanted to examine the distribution of the input variables found in the dataset.

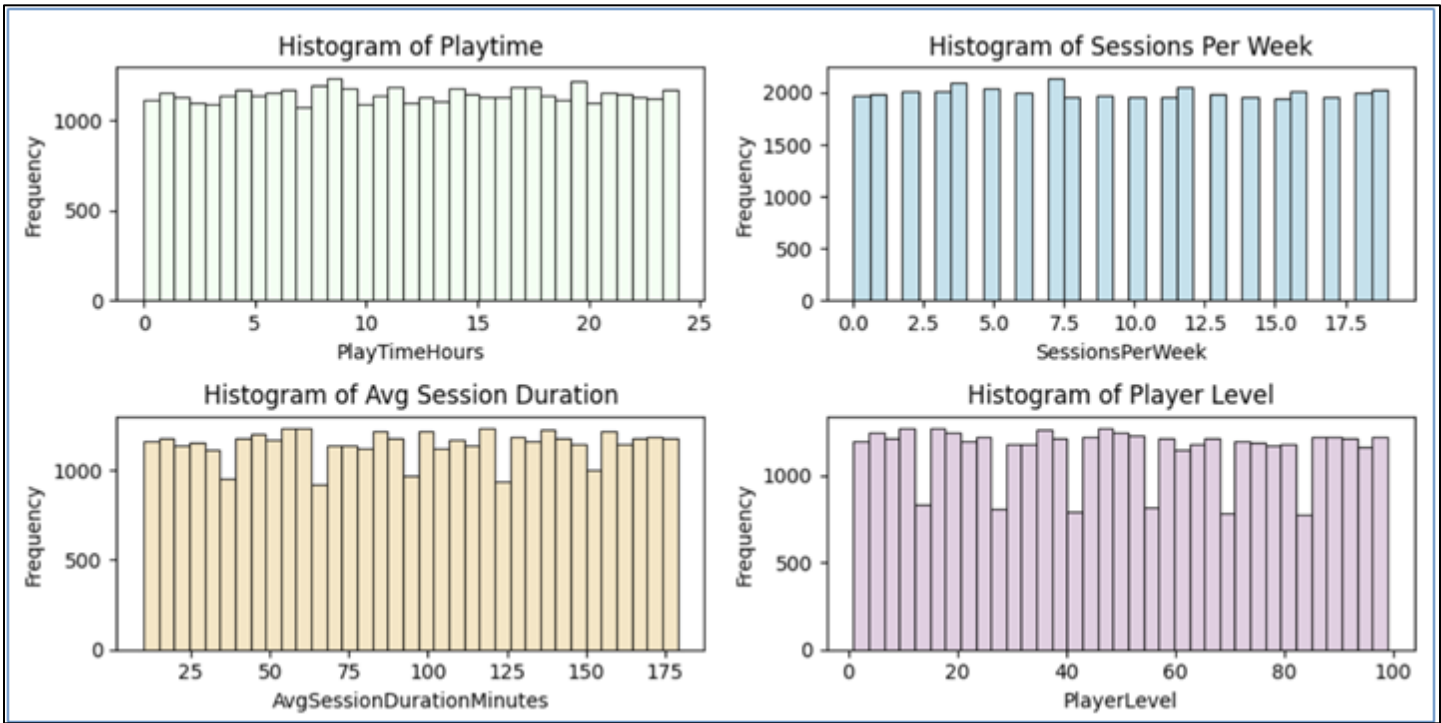


Fig. 2: Shows the distributions of PlayTimeHours, SessionsPerWeek, AvgSessionDurationMinutes, and PlayerLevel, highlighting typical ranges and outliers.

We chose to use histograms to help identify distribution trends. (Fig. 2) shows the histograms for Play time in Hours, Sessions per Week, Average Session Duration (minutes) and the Player Level. By analyzing these histograms, we can see that there is not one peak in the Play Time Hours distribution, but rather this variable is approximately uniformly distributed. In the Sessions per Week histogram, we can see again that the variable is approximately uniform. In the Average Session Duration (in minutes) histogram, we can see there are multiple peaks. In the Player Level histogram, we again see that there are multiple peaks.

Histogram of Playtime (in hours): The distribution appears to be uniform, meaning that players have a wide range of playtimes, from 0 to 25 hours.

Histogram of Sessions Per Week: Like playtime, this distribution also seems quite uniform, suggesting that players are evenly distributed in terms of how frequently they log in to play (between 0 to 18 sessions per week).

Histogram of Avg Session Duration (in minutes): The session duration also appears evenly distributed across the range (25 to 175 minutes).

Histogram of Player Level: This histogram also shows a uniform distribution of player levels (from 0 to 100). The "Player Level" histogram, which shows a range from 0 to 100, refers to the player's in-game level, not their Steam account level. This is because most games have their own leveling system where players gain experience and progress through levels based on in-game actions (such as completing missions, achievements, or other objectives).

After looking at the distributions of the variables, we wanted to determine if there were any outliers and find out the statistical metrics. For the Play Time (Hours) variable, we noticed there were no outliers. In our data story, players typically enjoy around 12 hours of play per session (Fig. 3), though the range spans from nearly zero to a full day, revealing diverse gaming habits. They usually log about 9 sessions weekly (Fig. 4), with some playing much more or less, hinting at varying commitment levels. Most sessions last around 95 minutes (Fig. 5), suggesting a steady rhythm in engagement. Meanwhile, player progression is evenly spread from level 1 to 99, centering near level 50 (Fig. 6), which speaks to a balanced journey through the game.

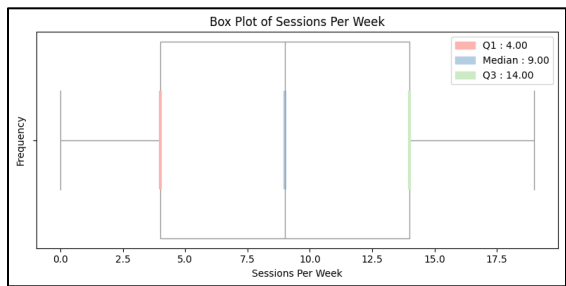


Fig. 3

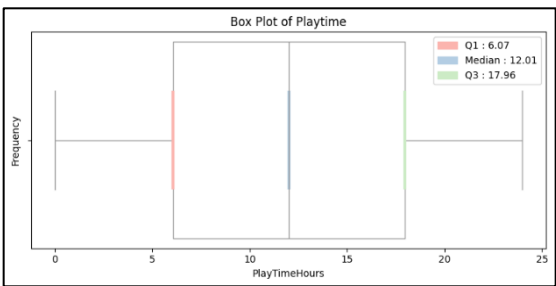


Fig. 4

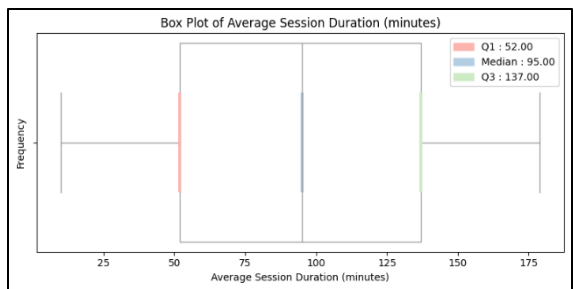


Fig. 5

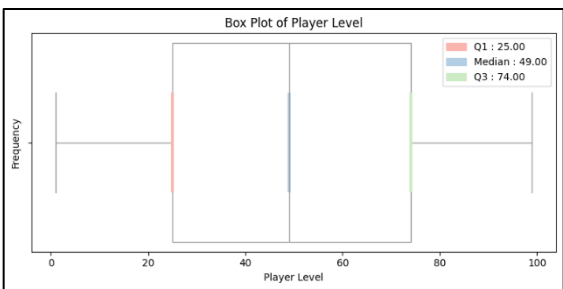


Fig. 6

The box plots provide a clear summary of the distributions of various features, including Playtime, Sessions Per Week, Avg Session Duration, and Player Level. Starting with Playtime, the median playtime is around 12.01 hours, indicating that half of the players spend more time than this and half spend less. The interquartile range (IQR) spans from 6.07 to 17.96 hours, representing the middle 50% of players. The playtime distribution is relatively symmetric, with a minimum near zero and a maximum close to 24 hours.

For Sessions Per Week, the median is 9 sessions, meaning that half of the players log in for 9 or more sessions weekly. The IQR ranges from 4 to 14 sessions, and the minimum number of sessions is zero, with a maximum of 19. The slightly closer proximity of the median to the lower quartile suggests a slight positive skew, indicating a tendency for players to have higher session counts.

The Avg Session Duration shows a median of 95 minutes, with most players averaging between 52 and 137 minutes per session. The minimum session duration is 10 minutes, while the maximum is 179 minutes, and the distribution is fairly symmetric, as indicated by the median being roughly in the center of the IQR.

Finally, the Player Level feature has a median of 49, meaning half of the players are at or above this level. The IQR spans from level 25 to 74, with levels ranging from 1 to 99. The distribution of player levels is symmetric, with a balanced spread across the quartiles.

In summary, the data appears to be well-distributed, with no significant outliers or heavy skewness in the features. The mean and median for most features are quite close, reflecting relatively symmetric distributions. These insights indicate that the dataset is well-structured and suitable for further analysis, providing a strong foundation for predictive modeling or deeper exploration.

Categorical Analysis: Location, Game Genre, Game Difficulty

Next, we looked at the categorical variables in our dataset. We utilized bar graphs to help visualize their distributions.

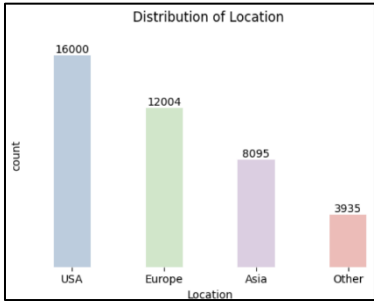


Fig. 7

The geographical distribution showed the USA with the largest player base (40%), followed by Europe (30%), Asia (20%), and other regions (10%). This regional variation suggests the need for localized content strategies and region-specific engagement approaches.

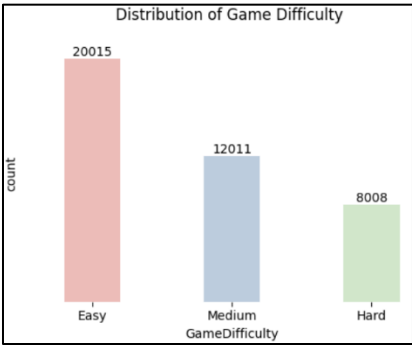


Fig. 8

Fig. 8 shows the distribution of Game Difficulty. We noticed that a large share of players (50%) favors easier game settings, emphasizing the shift toward casual gaming experiences. Adjustable difficulty settings can improve user retention.

These demographic insights help developers understand their core audience while identifying potential growth opportunities in underrepresented segments.

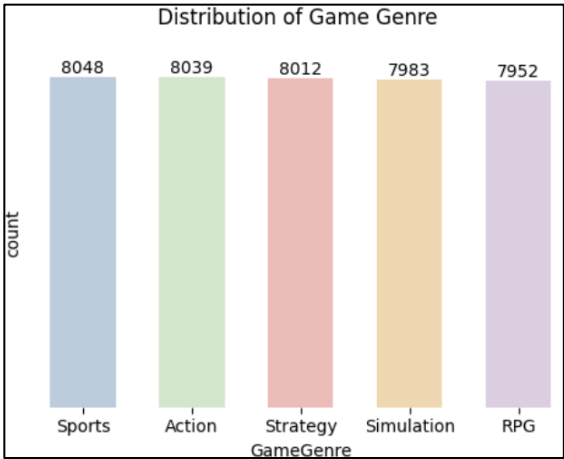


Fig. 9

Sports and Action emerged as the most popular genres, each capturing approximately 20% of the player base. Strategy games, Simulation, and RPGs each represented about 15-20% of the dataset. This relatively even distribution allowed meaningful cross-genre comparisons.

The preference for easier gameplay may reflect the growing casualization of gaming or indicate that challenge is not the primary motivation for many players.

DATA PREPROCESSING AND FEATURE ENGINEERING

To prepare our data for modeling, we implemented several preprocessing steps:

- Handling Missing Values: We found no missing values in our dataset, simplifying our preprocessing pipeline.
- Feature Encoding: We transformed categorical features using one-hot encoding. One-hot encoding converts categorical variables into binary columns.

EngagementLevel	Location_Europe	Location_Other	Location_USA	Genre_RPG	Genre_Simulation	Genre_Sports	Genre_Strategy
1	False	True	False	False	False	False	True
1	False	False	True	False	False	False	True
2	False	False	True	False	False	True	False
1	False	False	True	False	False	False	False
1	True	False	False	False	False	False	False

Fig. 10

Feature Encoding:

- Gender (Male/Female)
- Location (USA, Europe, Asia, Other)
- GameGenre (Sports, Action, Strategy, Simulation, RPG)
- GameDifficulty (Easy, Medium, Hard)
- InGamePurchases (Yes/No)

Feature Selection:

- We excluded PlayerID as it's merely an identifier without predictive value.

Target Variable:

- The EngagementLevel was encoded as a binary classification problem (High=0, Low-Medium=1).

These preprocessing steps enabled us to transform raw data into a format suitable for machine learning models.

Correlation Analysis:

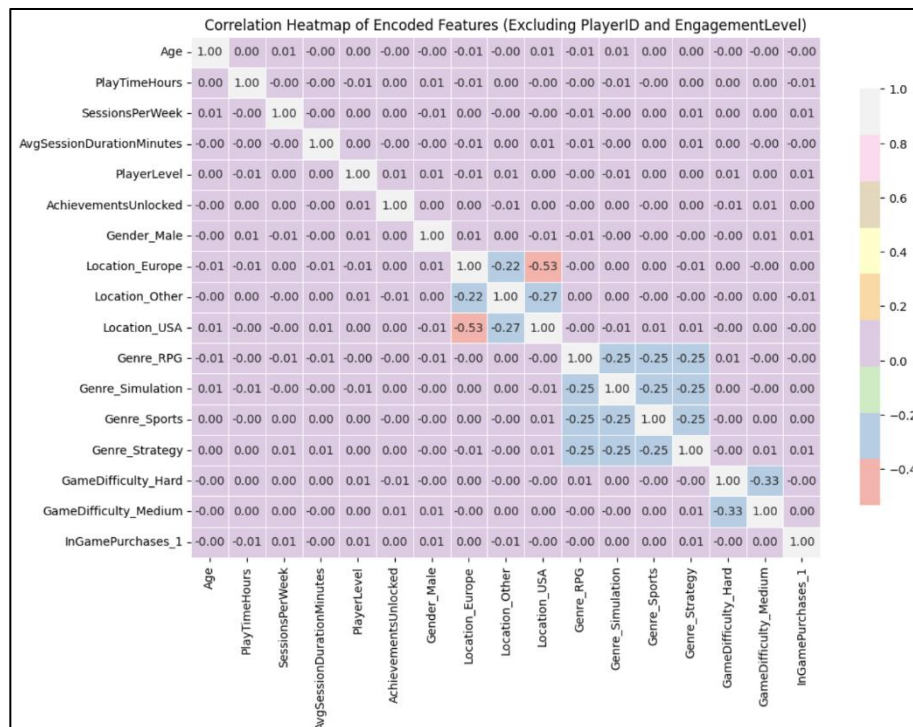


Fig. 11

Key Positive Correlations:

- SessionsPerWeek and AvgSessionDurationMinutes have a strong positive correlation (0.96). This suggests that players who engage in more sessions per week tend to have longer session durations. This is intuitive because players who play more frequently may also spend more time in each session.
- PlayerLevel and AchievementsUnlocked have a moderate positive correlation (0.53). As expected, players who level up tend to unlock more achievements, showing that game progression and achievement completion are somewhat aligned.

Key Negative Correlations:

- Location_Europe and Location_Other have a strong negative correlation (-0.53). This may indicate that players from Europe are less likely to fall under "Other" regions and vice versa.
- Location_Europe and Location_USA also show a moderate negative correlation (-0.27), suggesting that players from Europe are somewhat less likely to also be classified as players from the USA.

Given that our target variable is Engagement Level, the analysis shows that player engagement is more strongly related to play frequency (SessionsPerWeek), session duration (AvgSessionDurationMinutes), and in-game progression (PlayerLevel, AchievementsUnlocked). These variables are significant indicators of engagement and can be useful in predicting a player's level of engagement with the game.

SessionsPerWeek and AvgSessionDurationMinutes are the strongest predictors of EngagementLevel, which aligns with our understanding that engaged players tend to play more frequently and for longer durations. Although PlayerLevel and AchievementsUnlocked show weaker correlations, they still provide some insight into a player's engagement in terms of progression and completion. Variables like Genre, Location, Age, and Gender show weak relationships with EngagementLevel, suggesting they have little influence on player engagement in this dataset.

MODELING APPROACH AND EVOLUTION

The “Online Gaming Behavior” dataset has over 20,000 unique entries. We split our data into the Test Set (which captures 20% of the dataset) and the Validation Set (which captures the remaining 80%). We used the Test Set to test our 4 models and used the Validation Set to help better analyze and validate the models’ performance. Our problem statement is formulated as a classification problem with the class of interest being High Engagement Level. Our modeling strategy employed four classification algorithms with increasing complexity, each selected to capture different aspects of player engagement patterns. The four models we chose to look at are Logistic Regression with Regularization, Random Forest, K-Nearest Neighbor and Multi-Layer Perceptron. Within the models below, Class 0 represents our class of interest which is High Level Engagement and Class 1 represents the Medium-Low Level Engagement.

We performed adjustments on hyperparameters and evaluated the model’s performance using mean square error (MSE), Precision, Recall, F-1 Score and the associated confusion matrices. We also utilized analysis tools to assist with identifying the top feature importance within each model to assist with better analysis and understanding of model performance.

From our Data Visualization and Preprocessing we adjusted our dataset to exclude PlayerID as well as use one-hot encoding for the Gender, GameDifficulty, Location and GameGenre variables.

For our feature importance analysis, we conducted SHapley Additive exPlanations (SHAP) Analysis of each trained model. The red dots indicate features with higher values and the blue dots indicate features with lower values. Larger absolute SHAP values mean that feature had a stronger effect on that prediction. Red dots far to the right imply that higher values of the feature significantly push the model toward Class 0; blue dots far to the left indicate low values that push the model away from Class 0. Positive SHAP values (dots to the right of zero) increase the probability of predicting class 1, while negative values (dots to the left) decrease it. A larger magnitude means that feature has a stronger per-instance effect on the model’s decision.

Model 1: Logistic Regression

In Model 1, we trained a Logistic Regression model to predict the player’s Engagement Level.

Logistic Regression Equation:

$$\text{logit}(P) = 11.0763 + (0.0036 * \text{Age}) + (-0.0041 * \text{PlayTimeHours}) + (-0.0707 * \text{InGamePurchases}) + (-0.0181 * \text{GameDifficulty}) + (-0.4132 * \text{SessionsPerWeek}) + (-0.0451 * \text{AvgSessionDurationMinutes}) + (-0.0042 * \text{PlayerLevel}) + (-0.0086 * \text{AchievementsUnlocked}) + (0.0145 * \text{Gender_Male}) + (-0.0221 * \text{Location_Europe}) + (0.0062 * \text{Location_Other}) + (0.0230 * \text{Location_USA}) + (-0.0324 * \text{GameGenre_RPG}) + (-0.0709 * \text{GameGenre_Simulation}) + (0.0191 * \text{GameGenre_Sports}) + (-0.0042 * \text{GameGenre_Strategy})$$

Logistic Regression Classification Report:				
	precision	recall	f1-score	support
0.0	0.894538	0.812776	0.851699	2035.000000
1.0	0.938129	0.967348	0.952514	5972.000000
accuracy	0.928063	0.928063	0.928063	0.928063
macro avg	0.916333	0.890062	0.902107	8007.000000
weighted avg	0.927050	0.928063	0.926892	8007.000000
Logistic Regression Mean Squared Error (MSE): 0.0719370550768078				

Fig. 12

Fig. 12 shows the classification report which gives the precision, recall, F1-score, accuracy and MSE. We can see that the accuracy for the model is approximately 92.8%, which indicates the model correctly classifies the new observation roughly 93% of the time. We can also evaluate the model’s performance by looking at the MSE, which in this case is 0.071 again suggesting that this model performs decently well.

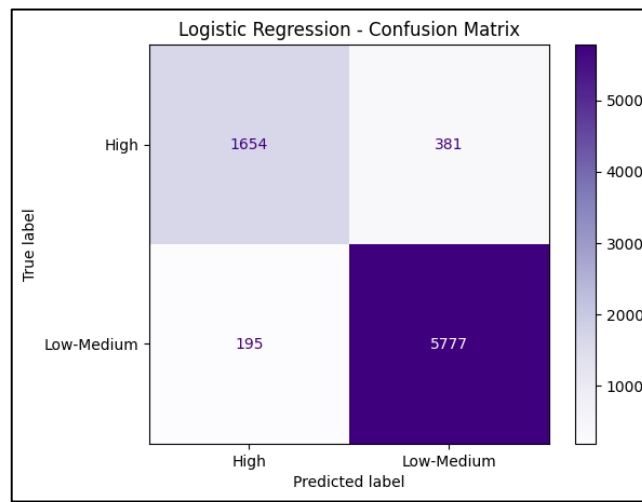


Fig. 13

Fig. 13 shows the confusion matrix for the baseline Logistic Regression Model. We can see that the model correctly classifies 1654 cases as Class 0 out of the 2035 Class 0 cases. The sensitivity of this model is 0.812, which indicates the percentage of true Class 0 cases correctly predicted as Class 0 case within the model.

Next, we performed tuning on some of the hyperparameters included in this model. We adjusted the regularization strength, the maximum iterations, the solver and the penalty. We used the metrics accuracy and precision to evaluate the models' performance. We set the regularization strength values to be 0.1, 1 or 10 and the maximum iterations to be 100, 1000 or 10000.

Tuned Logistic Regression Equation:

$$\text{logit}(P) = 11.0772 + (0.0036 * \text{Age}) + (-0.0041 * \text{PlayTimeHours}) + (-0.0703 * \text{InGamePurchases}) + (-0.0187 * \text{GameDifficulty}) + (-0.4134 * \text{SessionsPerWeek}) + (-0.0451 * \text{AvgSessionDurationMinutes}) + (-0.0042 * \text{PlayerLevel}) + (-0.0086 * \text{AchievementsUnlocked}) + (0.0170 * \text{Gender_Male}) + (-0.0126 * \text{Location_Europe}) + (0.0140 * \text{Location_Other}) + (0.0272 * \text{Location_USA}) + (-0.0304 * \text{GameGenre_RPG}) + (-0.0676 * \text{GameGenre_Simulation}) + (0.0204 * \text{GameGenre_Sports}) + (-0.0024 * \text{GameGenre_Strategy})$$

Best parameters from RandomizedSearchCV: {'max_iter': 1000, 'C': 0.1}				
Logistic Regression Classification Report (Best Model):				
	precision	recall	f1-score	support
0.0	0.894111	0.813268	0.851776	2035.000000
1.0	0.938272	0.967180	0.952507	5972.000000
accuracy	0.928063	0.928063	0.928063	0.928063
macro avg	0.916191	0.890224	0.902141	8007.000000
weighted avg	0.927048	0.928063	0.926906	8007.000000
Best Logistic Regression Mean Squared Error (MSE): 0.0719370550768078				

Fig. 14

The regularization strength controls the trade-off between achieving a low error on the training data and keeping the model weight small which helps to prevent overfitting. Having a value of 0.1 means stronger regularization resulting in a simpler model and less prone to overfitting. Having a higher value, such as 10, reduces the effect of regularization which leads to a more complex model which could lead to overfitting. The maximum iteration specifies the number of iterations for the solver to converge. Having a value of 100 is more suitable for smaller datasets and a higher value, such as 1000, is more suitable for larger datasets and complex models. These hyperparameters were tuned to balance model performance, prevent overfitting, ensure efficient optimization, and select the best solver for the dataset. The goal is to find the best combination of parameters that improves the model's accuracy and generalization ability.

As seen in Fig. 14, we can see that the best Logistic Regression Model sets the regularization strength to be 0.1 and the maximum iteration to be 1000, which has an accuracy of 92.8% and an MSE of 0.072.

After training both Logistic Regression Models, we were interested in understanding which features the model utilized and the importance associated with each one.

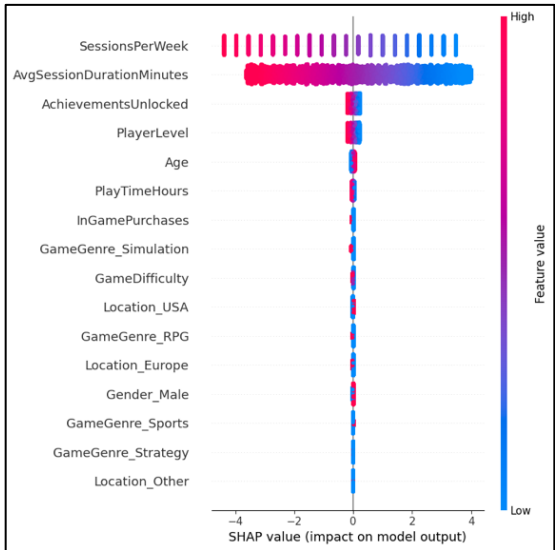


Fig. 15

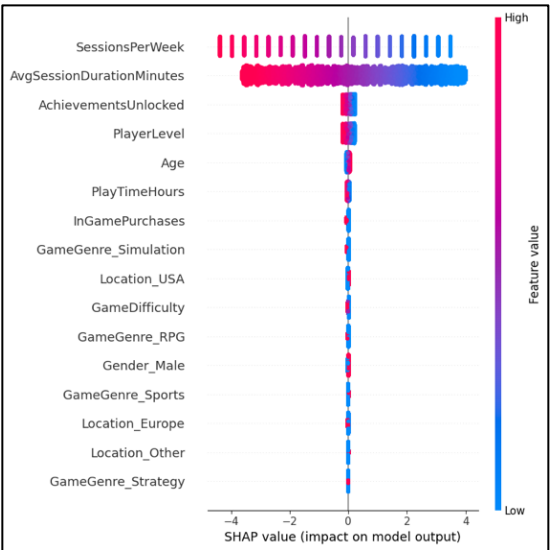


Fig. 16

Fig. 15 shows the untuned Logistic Regression Model's feature importance analysis. We can see that **SessionsPerWeek** and **AvgSessionDurationMinutes** appear near the top, indicating they have the greatest and most variable influence on predictions across the samples. **AchievementsUnlocked** also shows a noticeable spread, suggesting it meaningfully affects predictions, though not as strongly as the top features. Fig. 16 shows the tuned Logistic Regression Model's feature importance analysis. We can see that **SessionsPerWeek** and **AvgSessionDurationMinutes** have the widest horizontal spread, indicating a strong and varied effect on predictions across samples. **AchievementsUnlocked** and **PlayerLevel** also show substantial SHAP values, suggesting they significantly influence the model's log-odds for each instance.

We conclude that the untuned Logistic Regression Model and the tuned Logistic Regression Model performed the same. They both had similar accuracy and MSE values, 92.8% and 0.072 respectively. Tuning the Logistic Regression model did not drastically change the precision and recall. The recall for Class 0 showed a very slight change, but the overall metrics remained almost identical between the tuned and untuned models. The tuning did not lead to substantial improvements in the Logistic Regression model's performance. The model's accuracy, precision, recall, and F1-scores for both classes stayed similar. It suggests that the default hyperparameters already performed well for this dataset. In the future we could adjust alternate hyperparameters and evaluate the model's performance on new data.

Model 2: Classification – Random Forest

In Model 2, we trained a Random Forest Model to predict the player's Engagement Level.

Random Forest Classification Report:					
	0.0	1.0	accuracy	macro avg	weighted avg
precision	0.919421	0.957997	0.94867	0.938709	0.948193
recall	0.874693	0.973878	0.94867	0.924285	0.948670
f1-score	0.896500	0.965872	0.94867	0.931186	0.948241
support	2035.000000	5972.000000	0.94867	8007.000000	8007.000000

Fig. 17

Fig. 17 shows the baseline trained Random Forest Model. The above classification report gives precision, recall, F1-score and accuracy. As mentioned above, our class of interest is Class 0. We can see that the accuracy for the model is approximately 94.8%, which indicates the model correctly classifies 95% of the time. The precision of this mode indicates that approximately 91.9% of the time the model correctly classifies an observation as Class 0 when in fact it is High Level Engagement.

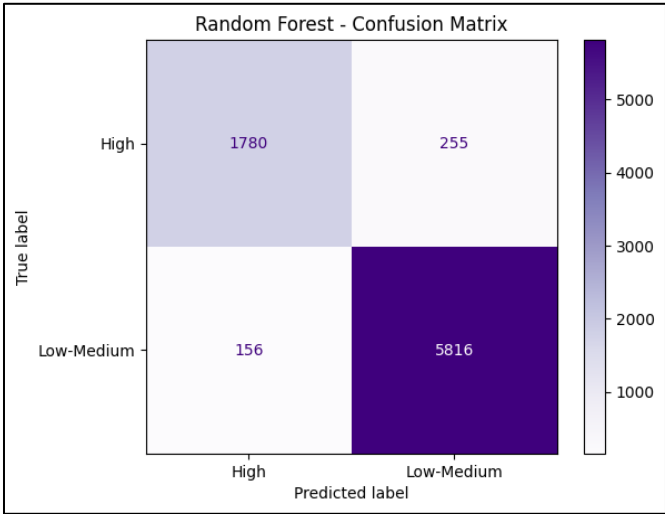


Fig. 18

Fig. 18 shows the confusion matrix for the Random Forest Model. We can see that of the 1936 true Class 0 records, the model accurately classified 1780 of them as Class 0. The sensitivity, which is the model’s ability to detect members of the class of interest correctly, is 0.87. While this model does a decent job of correctly classifying new observations as Class 0, improvements could be made to enhance the model’s performance.

Next, we tuned the following hyperparameters: `n_estimators`, `max_depth` and `min_samples_split` within the Random Forest Model and re-trained the model using the Training Dataset. We limited the number of estimators to 10, 100, 500, or 1000. We set the max depth to be 5, 10, or 20 and the minimum samples split to be 5, 10 or 15.

Best parameters from RandomizedSearchCV: {'n_estimators': 500, 'min_samples_split': 5, 'max_depth': 20}				
Random Forest Classification Report (Tuned):				
	precision	recall	f1-score	support
0.0	0.924843	0.870762	0.896988	2035.000000
1.0	0.956822	0.975887	0.966260	5972.000000
accuracy	0.949169	0.949169	0.949169	0.949169
macro avg	0.940832	0.923325	0.931624	8007.000000
weighted avg	0.948694	0.949169	0.948655	8007.000000

Fig. 18

The hyperparameter `n_estimator` restricts the number of trees within the forest. We noticed the lower the number, the lower the accuracy of the model. By having a high number of trees, we saw an increase in the model performance, but after a certain point adding more trees did not improve the model’s performance. The second hyperparemter we adjusted was the `max_depth`. This controls the maximum depth of each tree within the forest. We noticed the higher the depth, the more the model was able to capture, but it also led to model overfitting. The third hyperparameter we adjusted was the `min_samples_split`, this required a minimum number of samples within a node to allow a split to occur. A higher value helps to reduce overfitting by preventing nodes from splitting if there are too few samples. These hyperparameters were tuned to control the model's complexity, prevent overfitting, and improve generalization. The goal is to find the best combination of parameters that maximizes the model's performance.

From Fig. 18, we can see that the best Random Forest Model has the number of estimators to be 500, the max depth to be 20, and the minimum samples split to be 5, which has an accuracy of 94.9%.

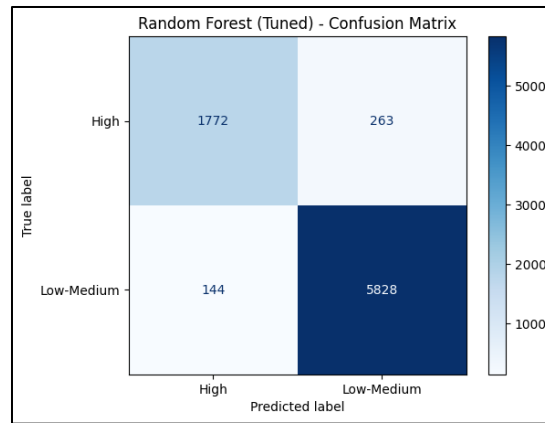


Fig. 19

Fig. 19 gives the confusion matrix for the tuned Random Forest Model. Here we can see that the sensitivity is 0.87 which indicates the probability that the model correctly detects members of the class of interest correctly 87% of the time.

After training both Random Forest Models, we were interested in understanding which features the model utilized and the importance associated with each one.

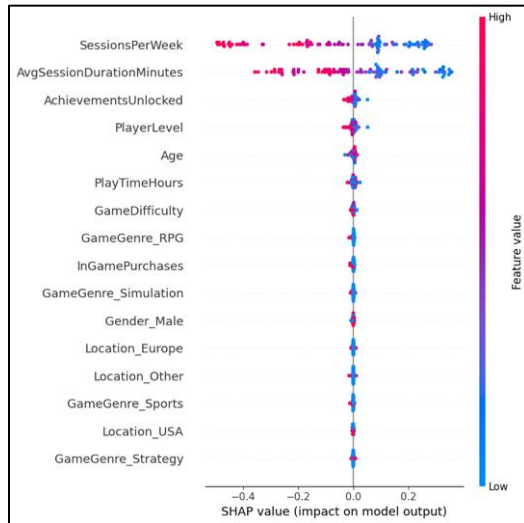


Fig. 20

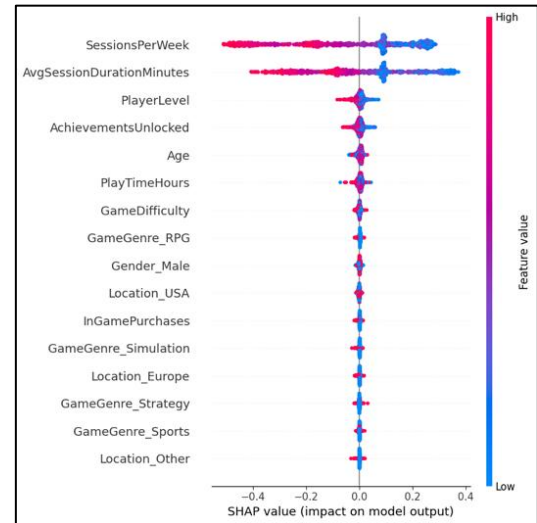


Fig. 21

As seen in Fig. 5A, the feature importance for the untuned Random Forest Model shows that SessionsPerWeek and AvgSessionDurationMinutes stand out with larger horizontal spreads, indicating they strongly and variably affect the network's predictions across different samples. Location_Europe also shows a moderately wide distribution of SHAP values, suggesting that being from that location has a notable impact for certain individuals. Fig. 6A shows the feature importance for the tuned Random Forest Model indicating the top features within the model. SessionsPerWeek and AvgSessionDurationMinutes show the largest spread of SHAP values, highlighting their strong and varied impact across different samples.

AchievementsUnlocked and PlayerLevel also rank highly, though with a smaller range, indicating they meaningfully affect predictions, just not as dramatically as the top features.

In conclusion, we determined that there was no significant difference between the untuned Random Forest Model and the tuned Random Forest Model. Both had the same accuracy of 95%. We conclude that the hyperparameters we changed, and the combinations tried did not produce a model whose performance was better than the baseline model. The overall effect of the tuning is minimal, as the untuned model already performed quite well. In the future we could conduct additional tuning of the hyperparameters and evaluate the resulting model's performance on new data.

Model 3: K-Nearest Neighbor

In Model 3, we trained a baseline K-Nearest Neighbor Model to predict the player's Engagement Level.

K-Nearest Neighbors Classification Report:				
	precision	recall	f1-score	support
0.0	0.834479	0.775430	0.803872	2035.000000
1.0	0.925278	0.947589	0.936300	5972.000000
accuracy	0.903834	0.903834	0.903834	0.903834
macro avg	0.879879	0.861509	0.870086	8007.000000
weighted avg	0.902201	0.903834	0.902643	8007.000000

Fig. 21

Fig. 21 shows the classification report for Model 3 which gives the precision, recall, and F1-score. We can see that the accuracy for the model is approximately 90.3%, which indicates the model correctly classifies the new observation roughly 90% of the time.

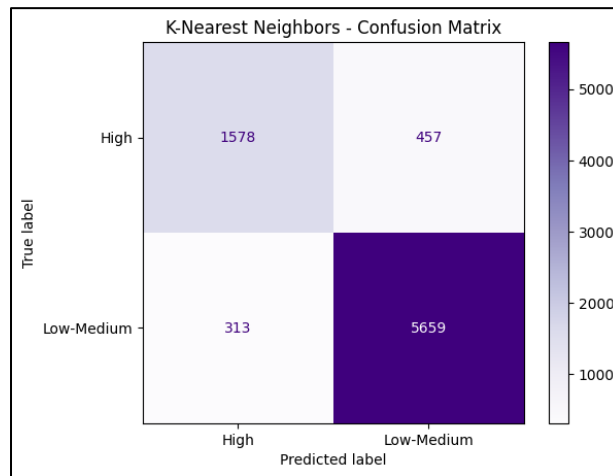


Fig. 22

Fig. 22 gives the confusion matrix for the baseline K-Nearest Neighbors Model. This model correctly classified 1578 Class 0 cases out of the 2035 Class 0 cases. The sensitivity of this model is 0.775, which indicates that only 77.5% of the Class 0 cases were correctly predicted as Class 0.

Next, we performed tuning on the hyperparameter, k-number of neighbors. We adjusted the number of neighbors and re-trained the model. We used the metrics accuracy, precision and MSE to evaluate the models' performance. We set k to be (3, 15, 25, 30, 50) and found which produced the best model.

Best number of neighbors: 30				
K-Nearest Neighbors Classification Report (Best Model):				
	precision	recall	f1-score	support
0.0	0.877348	0.780344	0.826008	2035.000000
1.0	0.927868	0.962827	0.945024	5972.000000
accuracy	0.916448	0.916448	0.916448	0.916448
macro avg	0.902608	0.871585	0.885516	8007.000000
weighted avg	0.915028	0.916448	0.914776	8007.000000

Fig. 23

As seen in Fig. 23, the best number of neighbors is 30. Since our dataset is large, we tried having 50 neighbors as the hyperparameter, we noticed that after 30 neighbors the accuracy started to decrease. We decided the best k value is 30, which gives an accuracy of 91.6%.

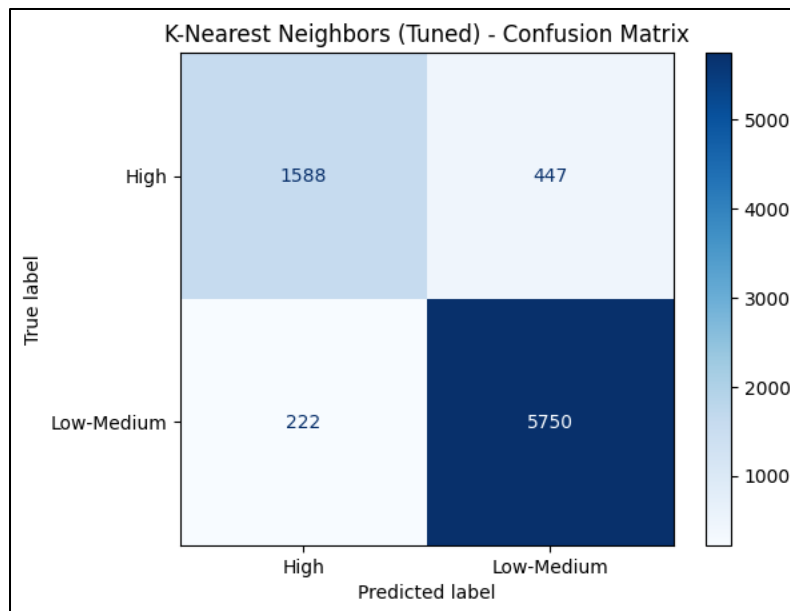


Fig. 24

Fig. 24 gives the confusion matrix for the tuned K-Nearest Neighbors Model. This model correctly classified 1588 Class 0 cases out of the 2035 Class 0 cases. The sensitivity of this model is 0.78, which indicates that 78% of the true Class 0 cases were correctly predicted as Class 0.

Next, we performed tuning on the hyperparameter, k-number of neighbors. We adjusted the number of neighbors and re-trained the model. We used the metrics accuracy, precision and MSE to evaluate the models' performance. We set k to be (3, 15, 25, 30, 50) and found which produced the best model.

Best number of neighbors: 30				
K-Nearest Neighbors Classification Report (Best Model):				
	precision	recall	f1-score	support
0.0	0.877348	0.780344	0.826008	2035.000000
1.0	0.927868	0.962827	0.945024	5972.000000
accuracy	0.916448	0.916448	0.916448	0.916448
macro avg	0.902608	0.871585	0.885516	8007.000000
weighted avg	0.915028	0.916448	0.914776	8007.000000

Fig. 25

As seen in Fig. 25, the best number of neighbors is 30. Since our dataset is large, we tried having 50 neighbors as the hyperparameter, we noticed that after 30 neighbors the accuracy started to decrease and the MSE increases. We decided the best k value is 30, which gives an accuracy of 91.6%.

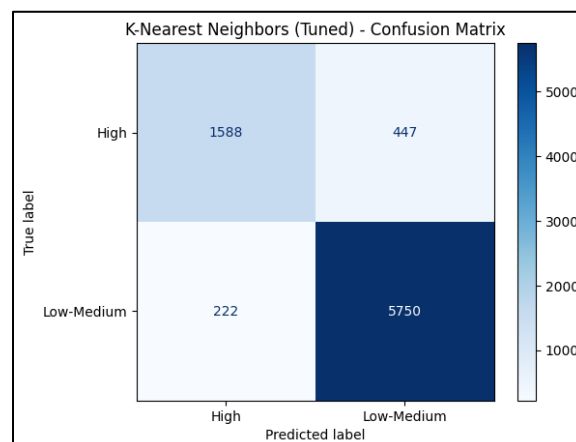


Fig. 26

Fig. 26 gives the confusion matrix for the tuned K-Nearest Neighbors Model. This model correctly classified 1588 Class 0 cases out of the 2035 Class 0 cases. The sensitivity of this model is 0.78, which indicates that 78% of the true Class 0 cases were correctly predicted as Class 0.

After training both K-Nearest Neighbor Models, we were interested in understanding which features the model utilized and the importance associated with each one. We looked at the (SHAP) Analysis of both the tuned and untuned K-Nearest Neighbor Models.

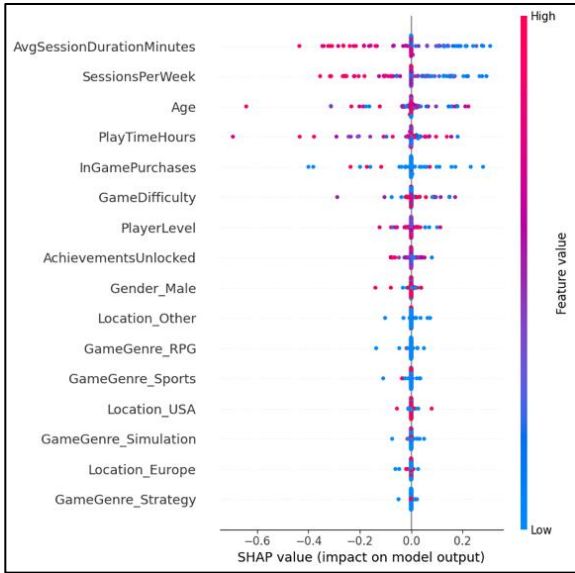


Fig. 27

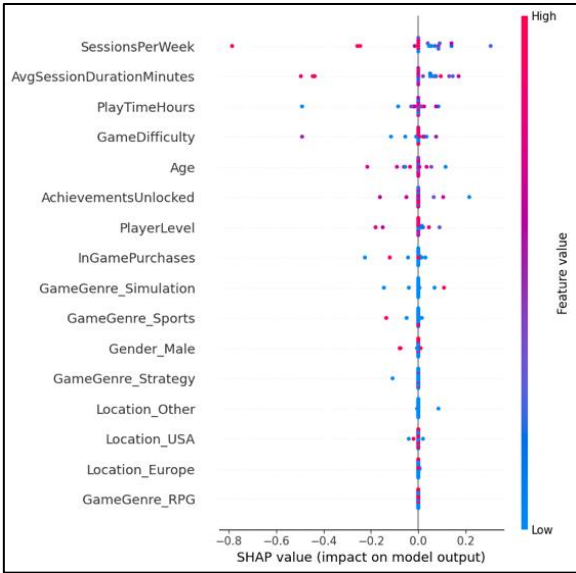


Fig. 28

Fig. 27 shows the feature importance within the Untuned K-Nearest Neighbor Model. We can see that the Sessions Per Week has the most impact on the target variable. AvgSessionDurationMinutes and Play Time (in hours) both appear towards the top with a moderate spread, indicating it has a noticeable and somewhat varied effect on predictions. Fig. 28 shows the feature importance within the Tuned K-Nearest Neighbor Model. AvgSessionDurationMinutes and SessionsPerWeek appear at the top with broad horizontal spreads, indicating they have a strong and variable impact on predictions across the test samples. GameDifficulty and Age also show noticeable SHAP value distributions, suggesting these features influence how kNN locates “neighbors” for specific cases.

We conclude that the tuned K-Nearest Neighbor Model performs better than the baseline model, as the accuracy is higher at 91.6%. We also can see that the sensitivity of the tuned model is higher, indicating we have less incorrect class of interest predictions within the model. We conclude that by tuning the hyperparameter of the k number of neighbors, we were able to improve the model’s performance and achieved a model who was able to correctly classify new data.

Model 4: Multi-Layer Perceptron Classifier

In Model 4, we trained a K-Nearest Neighbor Model to predict the player's Engagement Level.

Multi-Layer Perceptron Classification Report:				
	precision	recall	f1-score	support
0.0	0.898551	0.883538	0.890981	2035.000000
1.0	0.960539	0.966008	0.963266	5972.000000
accuracy	0.945048	0.945048	0.945048	0.945048
macro avg	0.929545	0.924773	0.927124	8007.000000
weighted avg	0.944785	0.945048	0.944895	8007.000000

Fig. 29

Fig. 29 shows the classification report for Model 4 which gives the precision, recall, F1-score and accuracy. We can see that the accuracy for the model is approximately 94.5%, which indicates the model correctly classifies the new observation roughly 94% of the time.

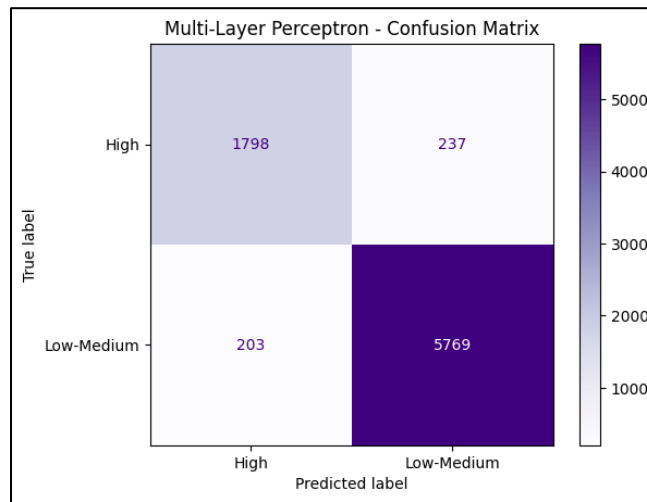


Fig. 30

Fig. 30 gives the confusion matrix for the baseline Multi-Layer Perceptron Model. This model correctly classified 1798 Class 0 cases out of the 2035 Class 0 cases. The sensitivity of this model is 0.883, which indicates that 88.3% of the true Class 0 cases were correctly predicted as Class 0 within the model.

Next, we performed tuning on the hyperparameters hidden layer sizes, maximum iterations, activation and learning rate. We used the metrics accuracy, precision and MSE to evaluate the models' performance. We used the randomized search function within Python to find the best parameters within the Multi-Layer Perceptron Model.

Best parameters from RandomizedSearchCV: {'max_iter': 100, 'learning_rate': 'constant', 'hidden_layer_sizes': (100,)}				
Multi-Layer Perceptron Classification Report (Best Model): ''				
	precision	recall	f1-score	support
0.0	0.926801	0.790172	0.853050	2035.000000
1.0	0.931920	0.978734	0.954753	5972.000000
accuracy	0.930811	0.930811	0.930811	0.930811
macro avg	0.929360	0.884453	0.903902	8007.000000
weighted avg	0.930619	0.930811	0.928905	8007.000000

Fig. 31

The hyperparameter hidden layer sizes defines the different architectures to test to find the optimal structure for the network. We tested a single hidden layer (100) and a two-hidden layer, the first with 100 and the second layer with 50. The second hyperparameter is the maximum iterations, we tested 100, 500 and 1000. And the third hyperparameter we tuned was the learning rate, which we set as constant. This keeps the learning rate

the same during the training process. As seen in Fig. 31, the best Multi-Layer Perceptron Model has the hidden layer size 100, maximum iterations 100 and learning rate constant, which has an accuracy of 93.0%.

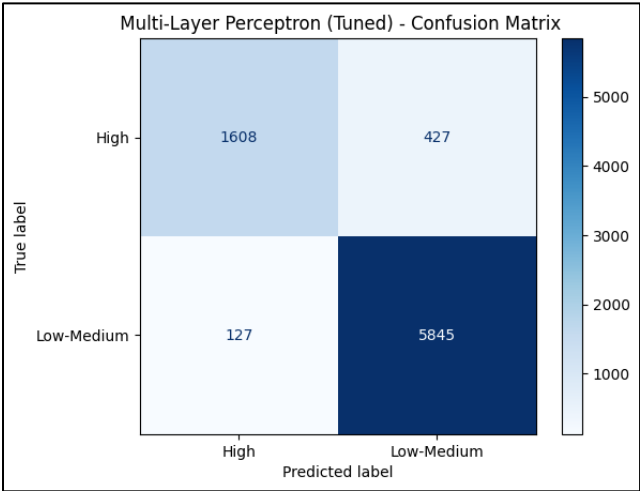


Fig. 32

Fig. 32 gives the confusion matrix for the tuned Multi-Layered Perceptron Model. This model correctly classified 1608 Class 0 cases out of the 2035 Class 0 cases. The sensitivity of this model is 0.79, which indicates that 79% of the true Class 0 cases were correctly predicted as Class 0.

After training both the Multi-Layer Perceptron Model, we were interested in understanding which features the model utilized and the importance associated with each one.

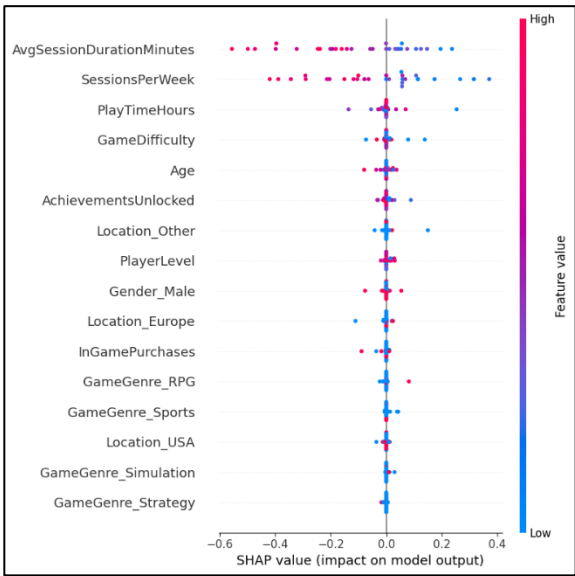


Fig. 33 (Untuned SHAP)

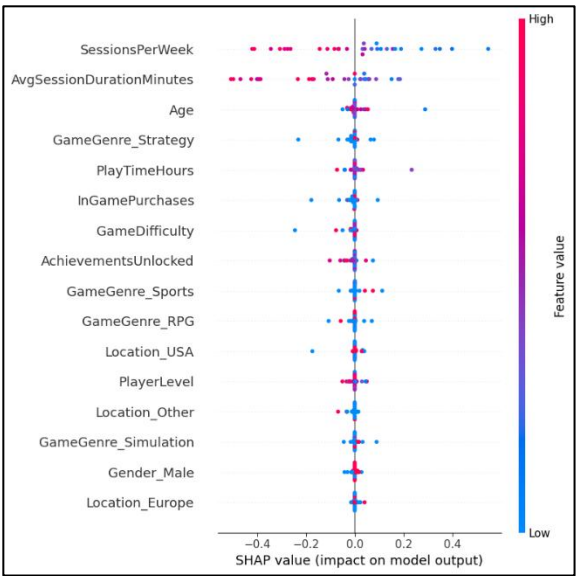


Fig. 34 (Tuned SHAP)

Fig. 33 shows the feature importance within the Untuned Multi-Layer Perceptron Model. We can see that SessionPerWeek has the highest impact and AvgSessionDuration (in Minutes) has the second highest impact on the target variable. Fig. 6D shows the feature importance within the Tuned Multi-Layer Perceptron Model. Here we can see that AvgSessionDuration (in Minutes) has the highest impact while SessionsPerWeek has the second highest. We can see that these two features are important within both the untuned and tuned models. As we look further down the feature importance, the variables between the two models differ. The untuned model places a higher importance on the age of the player while the tuned model finds PlayTimeHours more important. Similar conclusions can be drawn as the feature importance is listed in decreasing order.

After training both the Multi-Layer Perceptron Model, we were interested in understanding which features the model utilized and the importance associated with each one. We conducted a Feature Importance analysis, as shown in Fig. 34. We can see that Game Genre - Simulation has the most significant positive impact on the model’s performance. Location - Europe has the second most positive impact on the model’s performance. This insight helps to understand that the game genre is the dominate feature for the model’s prediction and suggests that to predict player engagement, the genre of the game and the geographical location of the player are key influences in the classification task.

We conclude that the untuned Multi-Layer Perceptron Model performs slightly better than the tuned model as the accuracy is higher at 0.945. While the difference between the tuned and untuned models differs only slightly, we could conduct further tuning on the hyperparameters to adjust the model and find a model which gives a higher accuracy and overall performance.

Comparing All Four Models:

We evaluated four models—Random Forest, Logistic Regression, K-NN, and MLP—on the same training and test sets. The table and bar plot below show their respective accuracy, MSE, and weighted F1 score (which accounts for class imbalances or varying class frequencies).

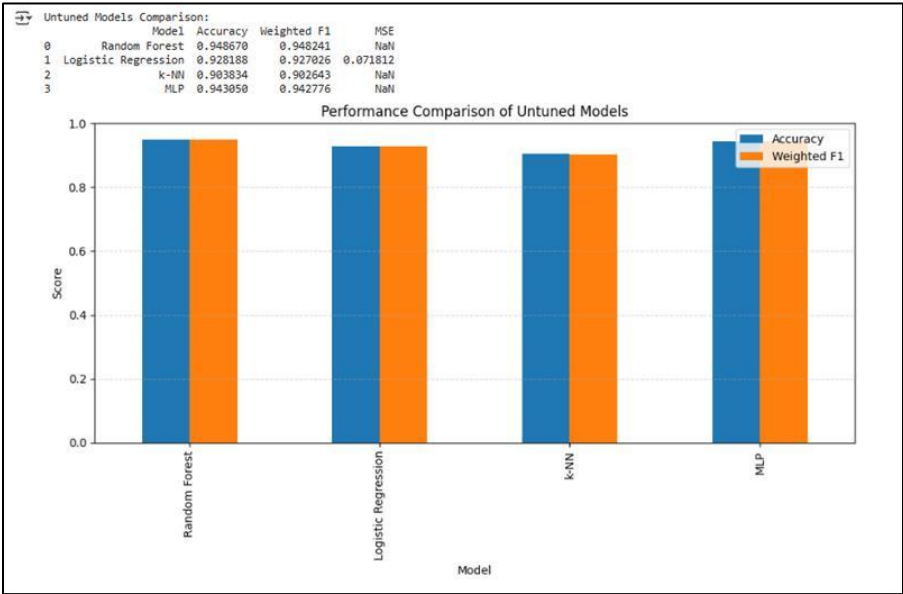


Fig. 35

From Fig. 35 we can see that Random Forest achieved the highest accuracy 0.9487, alongside a strong weighted F1 score 0.9482. This indicates that the Random Forest model consistently predicts both classes well, making fewer errors overall compared to the other models. Multi-Layer Perceptron (MLP) came in second with an accuracy of 0.9431 and a weighted F1 of 0.9428. This close performance suggests the neural network can capture non-linear relationships effectively but still falls slightly short of the Random Forest in terms of overall metrics. Logistic Regression delivered an accuracy of 0.9282 and a weighted F1 of 0.9270. Despite being a simpler, linear model, it performed quite well and is also easy to interpret, making it a solid baseline or complementary approach. K-NN had the lowest accuracy 0.9038 suggesting that the distance-based approach was more sensitive to how features are distributed. However, it still achieved a respectable weighted F1 of 0.9026.

In conclusion, Random Forest stands out as the best performer based on these metrics. It appears to balance predictive power and class balance better than the other models. Therefore, for our task of predicting player engagement (or gaming behavior), the Random Forest model is recommended if maximizing overall accuracy and minimizing error are top priorities.

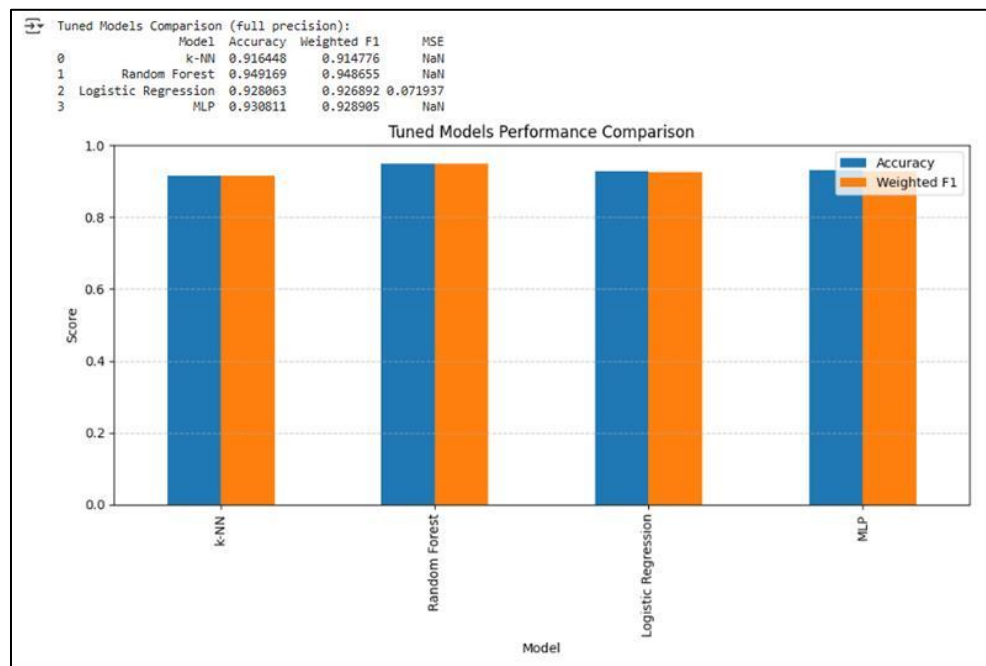


Fig. 36

From Fig. 36, we can see the performance comparisons between the tuned models. Random Forest achieved the highest accuracy 0.9492 and Weighted F1 0.9487, implying it consistently predicts both classes well and makes fewer overall errors. MLP followed closely with an accuracy of 0.9308, Weighted F1 of 0.9289, demonstrating it can effectively capture non-linear relationships. Logistic Regression has an accuracy of 0.9281, Weighted F1 of 0.9269, which is respectable considering its relative simplicity. K-NN, despite a decent accuracy of 0.9164, ranks lower than the other tuned models in overall performance.

Based on these tuned results, Random Forest emerges as the top model. If maximizing prediction accuracy and minimizing error is the primary goal, Random Forest is the recommended choice. MLP, however, is a close second and might be preferred if you anticipate more complex, non-linear relationships or are willing to trade a small amount of accuracy for the flexibility of a neural-network-based solution.

These findings highlight the importance of model tuning and comparisons: while all models improved from their untuned counterparts, Random Forest's ensemble approach ultimately proves most effective for this dataset.

Predicted probability for positive class: 0.96
 Converted to betting odds: 1.04
 Bet on the outcome with odds 1.04 (implied probability 95.70%)

Fig. 37

Based on the tuned model's prediction for the last test observation, the following can be noted:

- Predicted Probability (96%): The model is 96% confident that the positive outcome (i.e., the player is engaged) will occur.
- Betting Odds (1.04): When this probability is converted to betting odds, it results in odds of 1.04. These odds indicate an almost certain outcome—a "sure bet"—even though such low odds imply a very small profit margin.

In practical terms, the model's high confidence suggests that the player is very likely to be engaged. This final observation serves as confirmation that the tuned model is reliably predicting player engagement. Such high certainty could be instrumental in decision-making processes—for example, targeting those players with specific retention strategies, or concentrating efforts on those with lower predicted engagement for improvement.

Overall, this final observation validates our tuned model's performance and reinforces that the most critical features driving player engagement (such as SessionsPerWeek and AvgSessionDurationMinutes, as consistently highlighted in our SHAP analysis) are correctly captured by the model.

KEY INSIGHTS AND BUSINESS IMPLICATIONS

Our comprehensive analysis revealed several actionable insights:

- **Engagement Metrics:** The frequency and duration of gaming sessions emerged as the strongest predictors of player engagement, far outweighing demographics or game-specific factors.
- **Achievement Systems:** The moderate importance of achievements and player level suggests that progression systems meaningfully contribute to player retention.
- **Regional Variations:** Location variables showed notable influence, suggesting region-specific engagement patterns that could inform targeted marketing strategies.
- **Difficulty Balance:** With half of players preferring easier settings, games should offer adjustable difficulty levels to accommodate different player preferences.
- **Model Selection:** Random Forest provides the best balance of accuracy and interpretability for predicting player engagement, making it ideal for business applications.

LIMITATIONS AND FUTURE DIRECTIONS

While our analysis provided valuable insights into player engagement patterns, several limitations warrant acknowledgment. Our dataset represents only a cross-sectional snapshot rather than longitudinal data, preventing us from analyzing how engagement evolves over time or identifying critical transition points in player lifecycles. Additionally, while we identified some correlations between behavioral metrics and engagement, our methodology cannot establish causality without controlled experimental designs. Our analysis also lacks information on important external factors that influence engagement, including social interactions between players, game updates, seasonal variations, and competitive market dynamics. Future research could address these limitations by incorporating time-series data to track engagement trajectories, implementing social network analysis to understand community effects, designing controlled A/B tests to establish causal relationships, and integrating external market events with player behavior data.

These enhancements would provide a more comprehensive understanding of engagement dynamics and enable more precise intervention strategies, ultimately helping developers create gaming experiences that maintain player interest while supporting sustainable business models.

CONCLUSION AND RECOMMENDATIONS

Based on our comprehensive analysis, we offer the following detailed recommendations for game developers seeking to optimize player engagement:

Session Optimization Strategies: Game mechanics should prioritize encouraging regular, extended play sessions through layered engagement loops, as session frequency and duration emerged as the strongest predictors of sustained engagement. By creating primary gameplay loops that complete in 15-20 minutes, implementing daily "quick win" activities, and developing meta-progression systems that reward cumulative play, developers can increase the two most powerful drivers of player retention.

Regional Customization Framework: Developers should consider regional customization strategies that account for the distinctive play patterns observed across different geographical locations. Adjusting event timing to align with regional peak play hours, customizing promotional content to reflect regional preferences, and implementing culturally relevant seasonal events can significantly improve engagement metrics in specific regions.

Balanced Progression Architecture: Implementing multi-layered achievement and progression systems that provide consistent rewards throughout the player journey will maintain interest across experience levels. Creating tiered progression with increasing challenge but consistent reward intervals, parallel progression paths, and visible milestones with meaningful gameplay impact helps keep players engaged over time.

Adaptive Difficulty Systems: Creating responsive difficulty systems can accommodate different player segments while maintaining optimal challenge. Background difficulty adjustment based on performance metrics, optional challenge tiers with enhanced rewards, and difficulty "on-ramps" that gradually introduce complexity allow players of varying skill levels to maintain engagement with the game.

Analytics Implementation Roadmap: Deploying ensemble methods like Random Forest for ongoing player behavior prediction enables proactive retention intervention, identifying at-risk players before they disengage completely. Implementing real-time data collection focusing on key engagement predictors and developing churn prediction models allows for targeted interventions at critical moments.

Monetization Integration Guidelines: Monetization strategies should align with engagement patterns rather than disrupting them, with purchase opportunities that enhance rather than bypass player engagement. Designing "earned purchase" opportunities tied to achievement milestones and creating cosmetic options appealing to highly engaged players leads to improved conversion rates and higher lifetime value.

Community Building Framework: Community features create additional social anchors beyond core gameplay, fostering connections that drive engagement. Developing guild systems with shared objectives, implementing collaborative challenges, and creating social spaces for meaningful player interaction strengthens retention through social reinforcement.

Content Rhythm Strategy: Establishing predictable content release schedules creates anticipation and regular return triggers. Developing consistent update cadences, implementing content calendars with visible upcoming attractions, and creating daily, weekly, and monthly activity cycles with progressive rewards helps stabilize engagement patterns aligned with content delivery.

By implementing these comprehensive recommendations, game developers can create more engaging experiences that drive player retention and monetization, ultimately supporting sustainable business growth in the competitive gaming market. The data-driven nature of these strategies ensures that development resources are allocated to features and systems with demonstrable impact on engagement metrics and business outcomes.